Mid svp answers

1q. Explain digital video fundamentals. Discuss frames fields, frame rates, resolution, and chroma subsampling.

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1. Frames and Fields

These represent the temporal "snapshots" of a video.

- **Frames:** A frame is a single, complete static image. In **Progressive Scanning**, the entire frame is captured and displayed line by line from top to bottom at once.
- **Fields:** Used in **Interlaced Scanning**, a frame is divided into two fields: one containing the odd-numbered lines and the other containing even-numbered lines. This was originally designed to save bandwidth in analog TV while maintaining a high "perceived" refresh rate.

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2. Frame Rate (Temporal Resolution)

This is the number of frames or fields displayed per second, measured in **Frames Per Second (fps)**.

- **Standard Rates:** 24 fps (cinema), 30 fps (NTSC TV), and 60 fps (high-motion sports/gaming).
- **Significance:** Higher frame rates result in smoother motion. If the frame rate is too low, the human eye perceives "flicker" or jerky movement.

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3. Resolution (Spatial Resolution)

Resolution refers to the number of pixels contained in a single frame, usually expressed as **Width × Height**.

- **Examples:** Full HD (1920×1080) or 4K UHD (3840×2160).
- **Impact:** Higher resolution allows for more fine detail but increases the amount of data that needs to be processed, stored, and transmitted.

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4. Chroma Subsampling

This is a critical technique in video processing that reduces the data size by taking advantage of the **Human Visual System (HVS)**. The HVS is much more sensitive to brightness (**Luminance**) than to color (**Chrominance**).

Video is split into Y (Luminance) and $CbCr$ (Chrominance). We keep all Y data but "downsample" the Cb and Cr data.

- **4:4:4:** No subsampling. Every pixel has its own color data.
- **4:2:2:** Half the horizontal color resolution (used in high-end production).
- **4:2:0:** The most common format (used in DVDs, Blue-rays, and web streaming). It uses one color sample for every 2×2 block of pixels, reducing the color data significantly without a massive loss in perceived quality.

Summary for Examination Purposes:

Feature	**Description**	**Key Metric**
Frame	Complete image at a point in time.	Unit: Frame
Field	Half an image (odd or even lines).	Used in Interlaced video
Frame Rate	Speed of image sequence.	Unit: fps
Resolution	Number of pixels in an image.	Unit: Pixels ($W \times H$)
Chroma Subsampling	Reducing color data to save bandwidth.	Format: J:a:b ratio

4 lesson

Question 3: Explain Block Matching Motion Estimation (BMME)

Block Matching Motion Estimation is the cornerstone of modern video compression (like H.264 and HEVC). Its primary goal is to reduce **temporal redundancy**—the fact that pixels in one frame are likely to appear in a slightly different position in the next frame.

1. The Core Methodology

Instead of analyzing every single pixel (which would be computationally impossible), the "Current Frame" is partitioned into a grid of non-overlapping rectangular blocks, typically called **Macroblocks** (e.g., 8×8 or 16×16 pixels).

2. The Step-by-Step Process

1. **Target Block Selection:** Pick a macroblock in the current frame (F_t).
2. **Search Window Definition:** Because objects usually don't teleport across the screen, the algorithm defines a **Search Window** in the "Reference Frame" (F_{t-1}). This window is centered at the original coordinates of the target block.
3. **The Search:** The algorithm slides the target block around the search window in the reference frame, pixel by pixel, looking for the best match.
4. **Vector Generation:** Once the best match is found, the displacement from the original (x, y) coordinates to the new (x', y') coordinates is recorded as a **Motion Vector (MV)**.

3. Matching Criteria (The Math)

To determine which block is the "best" match, we use error-measuring functions:

- **SAD (Sum of Absolute Differences):** The simplest and most popular. It adds up the absolute differences between corresponding pixels in the two blocks.
  - Formula: $SAD = \sum |Pixel_{Current} - Pixel_{Reference}|$
- **MSE (Mean Squared Error):** Squaring the differences. It is more accurate but requires more processing power.
- **MAE (Mean Absolute Error):** Similar to SAD but averaged over the number of pixels.

4. Limitations

Block matching assumes **translational motion** (moving left, right, up, down). It struggles with rotation, zooming, or objects that change shape (deformable motion).

Question 7: Explain Motion Compensation in Video Coding

If Motion Estimation is the "detective" work (finding where things went), **Motion Compensation** is the "architect" that uses that information to rebuild the frame using as little data as possible.

1. The Principle of Inter-frame Coding

In a standard video, 90% of the background often stays the same between two frames. Motion Compensation allows the encoder to describe a new frame by referring to a previous one, rather than sending a whole new set of pixels.

2. The Workflow of Compensation

- **Reference Frame Storage:** The encoder and decoder both keep a copy of the previously decoded frame in a "Frame Buffer."
- **Prediction (The "Shift"):** Using the **Motion Vectors** sent by the encoder, the decoder takes blocks from the Reference Frame and shifts them to their new positions. This creates a **Predicted Frame** ($\hat{F}_t$).
- **Residual (Error) Calculation:** The Predicted Frame isn't perfect (because of lighting changes or non-linear motion). The encoder subtracts the Predicted Frame from the Actual Frame to get the **Residual Frame**:
  - $Residual = Actual \ Frame - Predicted \ Frame$
- **Encoding the Residual:** This residual image is usually mostly black/empty. It is then compressed using **DCT (Discrete Cosine Transform)** and sent to the decoder.

3. Why it provides High Efficiency

- **Data Savings:** Sending a small Motion Vector (a few bits) and a sparse Residual Frame is much "cheaper" than sending a full 1920×1080 pixel array.
- **Bitrate Control:** By adjusting how much detail we keep in the Residual, we can control the quality and size of the video file.

4. Types of Frames in Compensation

- **I-Frames (Intra):** No compensation. They are full images (like JPEGs).
- **P-Frames (Predicted):** Use motion compensation from a *previous* frame.
- **B-Frames (Bi-predictive):** Use motion compensation from both *previous and future* frames for even higher efficiency.

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5 lesson

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1. Explain image segmentation techniques: threshold, edge-based, region-based, and clustering-based methods.

Image segmentation is the process of partitioning a digital image into multiple segments (sets of pixels) to make the representation of an image more meaningful and easier to analyze.

A. Thresholding Methods

This is the simplest form of segmentation. It turns a grayscale image into a binary image based on a "threshold" value.

- **How it works:** If a pixel's intensity is higher than the threshold, it's assigned to the "object" (white); otherwise, it's "background" (black).
- **Types: Global thresholding** uses one value for the whole image, while **Adaptive thresholding** changes the value based on local pixel neighborhoods to handle uneven lighting.

B. Edge-Based Methods

These methods look for sharp contrasts in intensity, which usually indicate the boundaries (edges) of objects.

- **How it works:** Filters like **Sobel**, **Prewitt**, or **Canny** edge detectors are applied to find discontinuities in brightness.
- **Challenge:** It often results in "broken" edges that need to be linked together to form a solid boundary.

C. Region-Based Methods

Instead of looking for edges, these methods look for groups of similar pixels.

- **Region Growing:** Starts with a "seed" pixel and adds neighboring pixels that have similar properties (like color or intensity).
- **Region Splitting and Merging:** The image is subdivided into quadrants until all pixels in a quadrant are similar. If adjacent quadrants are similar, they are merged back together.

D. Clustering-Based Methods

These treat segmentation as a data science problem, grouping pixels into "clusters" based on their features (color, texture, or position).

- **K-Means Clustering:** A popular algorithm where pixels are assigned to one of K clusters by minimizing the distance between the pixel and the cluster center (centroid). It is excellent for multi-colored images.

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8. Explain change detection in video. Compare frame differencing, running average, and Gaussian background modeling approaches.

Change detection identifies moving objects by comparing the current video frame against a reference "background" model.

A. Frame Differencing

This is the most basic approach for detecting motion.

- **Mechanism:** It subtracts the previous frame from the current frame: $D(t) = |I(t) - I(t-1)|$. If the difference is greater than a threshold, motion is detected.

- **Pros:** Extremely fast and computationally "cheap."
- **Cons:** It only detects the edges of moving objects and fails if the object stops moving (it disappears from the difference map).

B. Running Average (Temporal Averaging)

This creates a more stable background model over time.

- **Mechanism:** The background is updated gradually: $B(t) = (1-\alpha)B(t-1) + \alpha I(t)$, where α is a small learning rate.
- **Pros:** It can handle slow changes in lighting (like the sun moving).
- **Cons:** If an object stays still for a while, it eventually "fades" into the background model (the "ghosting" effect).

C. Gaussian Background Modeling (MoG)

This is a sophisticated statistical approach often called **Mixture of Gaussians (MoG)**.

- **Mechanism:** Instead of one color, each pixel's history is modeled as a distribution (a "bell curve"). If a new pixel value falls outside this curve, it is labeled as "foreground" (change).
- **Pros:** It is excellent at handling "noisy" backgrounds like waving trees or rippling water, which would trigger false alarms in simpler methods.
- **Cons:** Requires significant memory and processing power.

Comparison Table

Method	**Complexity**	**Handling Lighting Changes**	**Noise Tolerance**
Frame Differencing	Very Low	Poor	Low
Running Average	Medium	Moderate	Moderate
Gaussian (MoG)	High	Excellent	High